

Gesture Controlled Arm

Classical hand-gesture recogniser, 6-DOF DH arm with LM inverse kinematics and smooth motion

AT A GLANCE

95.6 %

Gesture accuracy (360 frames)

0.010 mm

IK round-trip max error

200/200

IK converged (position)

81 %

Jerk reduction (filters)

97 %

Pointing-axis accuracy

7.6 ms/frame

Recogniser speed

17/17

Unit tests

VERDICT

SIM PASS

TESTS 17/17

IK < 1 mm

DECLARED SUBSTITUTION

Eswar Mahalingam

Robotics & Automation Intern
Ghaziabad, UP, India
+91-9360548243
eswarmba05313@gmail.com
linkedin.com/in/eswar-mahalingam

1. Objective and brief mapping

The brief asks for a MediaPipe-free gesture recogniser evaluated on at least 300 synthetic frames with a confusion matrix, a 5/6-DOF arm with DH parameters, FK, numeric IK (damped least squares / LM with joint limits and a null-space posture task) plus analytic IK for the planar sub-chain, smooth motion (One-Euro filtering, velocity- and acceleration-limited joint trajectories, dead-zone), a scripted gesture-to-motion GIF, jerk plots before/after filtering, and tests on IK accuracy, joint limits, recogniser accuracy and jerk reduction. All of it is in **src/gesture_arm/** (numpy, OpenCV, matplotlib) and exercised by **sim/** and **tests/**; the MediaPipe adapter, webcam demo, Webots world and ROS 2 package are provided but declared not run here.

2. Architecture

```
camera frame (320x240 BGR, synthetic: render_hand -> hidden label)
-> YCrCb inRange(Cr 133-188, Cb 77-133) -> open/close 3x3 -> median -> largest
    contour (>= 600 px)
-> distance transform -> palm centre + radius R
-> convex hull -> convexity defects: depth > 0.45 R, angle < 95 deg, both ends > 1.75
    R => valleys
-> hull points > 1.85 R from palm, clustered, forearm rejected => fingertips
-> fingers = valleys + 1 (or 1 / 0 from tips); pointing direction; hand size
GestureArmLoop (15 Hz): 3-frame majority vote -> palm STOP / fist GRIP / point JOG(axis)
    / two POSITION(palm x,y -> arm y,x ; hand size -> z) / three TOOL_TOGGLE / four HOME
-> DeadZone 4 mm -> OneEuro(1.2 Hz, beta 0.4) -> ik_dls (LM, limits, null-space
    posture)
-> LimitedJointTrajectory (v <= 90..180 deg/s, a <= 300..600 deg/s^2) -> servo angles
Arm: 6-DOF spherical-wrist DH chain d1 0.10 a2 0.15 d4 0.14 d6 0.11 (m)
```

3. Implementation details

■ **Synthetic hands.** A palm ellipse, a short forearm stub, five capsule fingers (index/middle/ring/little/thumb extended in that order, folded fingers drawn as knuckle bumps so a fist is bulkier than the palm), five Indian skin tones, random rotation over 360 deg, scale 0.7-1.25, position anywhere in frame, a colour gradient background kept outside the skin chroma band, 3-8 clutter shapes, small skin-like distractors, lighting gain 0.75-1.2, sensor noise and blur. The label carries finger count, centre, palm radius and the index-finger direction.

■ **Recogniser.** The palm centre and radius come from the distance transform of the blob (largest inscribed circle), which is far more stable than the contour moments when fingers or the forearm are present. Valleys between fingers are convexity defects whose depth exceeds 0.45 R and whose two hull ends are both finger-length away from the palm - that single rule removed the false valleys created by the folded thumb and the wrist. Fingertips are hull points beyond 1.85 R, clustered at 0.9 R; a lone tip shorter than 2.05 R is a knuckle, and of two tips the wide one (mask width probed at 1.5 R) is the forearm.

■ **Arm.** Standard DH with the forearm length carried in d4 along the roll axis so J4-J5-J6 form a spherical wrist (verified: tool roll never moves the tip). Joint limits -150..150 deg on the base and hobby-servo ranges elsewhere. The geometric Jacobian is checked against central finite differences to 1e-6.

■ **IK.** Levenberg-Marquardt on the task error [position; 0.3 x rotation-vector]: $dq = J^T (J J^T + \lambda I)^{-1} e$, a step is accepted only if it lowers $|e|$ (else $\lambda \times 4$, step halved), joints sitting on a limit and pushed outward are removed from the Jacobian so the others take over, and a null-space term pulls towards the home posture. `ik_multistart` adds informed seeds (base yaw aimed at the target, the analytic elbow-up/down solutions) and random restarts. The analytic solver handles yaw + shoulder + elbow + wrist pitch with the tool held at a fixed world pitch (law of cosines on the wrist centre).

■ **Smooth motion.** DeadZone (4 mm) kills tremor, the One-Euro filter (min cutoff 1.2 Hz, beta 0.4) removes jitter without adding lag during fast moves, and LimitedJointTrajectory re-plans a discrete trapezoidal profile every tick with per-joint velocity and acceleration caps (the discrete stopping speed $v = -a \, dt/2 + \sqrt{(a \, dt/2)^2 + 2 \, a \, |d|}$ keeps the acceleration bound exact, which the naive continuous formula does not).

4. Verification

All 17 tests pass (sim/test_log.txt, 17 ok). The recogniser confusion matrix on 360 fresh frames (60 per class, seed 0) is below; every error is an off-by-one finger count, mostly four read as three when the little finger valley is shallow at small scale. Centroid error was 1.2 px mean / 2.7 px p95 and the detected palm radius was 0.95 +/- 0.01 of the rendered one, so position and depth mapping are usable. Pointing axis (4-way quantised) was right on 97 % of correctly counted point frames; the misses lie within a few degrees of a 45 deg boundary.

Group	What is checked	Tests	Result
Kinematics	zero-pose geometry; spherical wrist; Jacobian vs finite differences; 200-pose position round trip < 1 mm (max 0.010 mm); full-pose convergence >= 80 %; joint limits on unreachable targets; analytic planar IK exact and tool pointing down	7	PASS
Recogniser	skin mask isolates the hand; >= 90 % on 300 frames with >= 75 % recall per class; centroid/size/pointing axis on a known frame; no-hand frame rejected; MediaPipe landmark adapter counts fingers	5	PASS

KEY TAKEAWAY A YCrCb + convex-hull recogniser reached 95.6 % on 360 synthetic frames and drove a 6-DOF DH arm through LM inverse kinematics (max round-trip error 0.010 mm); dead-zone + One-Euro + velocity/acceleration limits cut joint jerk by 81 %.

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Smoothing	One-Euro halves jerk on a noisy moving signal while tracking within 6 mm; exponential + dead-zone semantics; limited trajectory respects v/a caps and reaches target	3	PASS
Control loop	palm mapping stays inside the workspace box; 75-frame tracking run within limits and < 5 mm final error	2	PASS

truth \ predicted	fist	point	two	three	four	palm	recall
fist	59	1	0	0	0	0	98 %
point	0	59	1	0	0	0	98 %
two	0	0	59	1	0	0	98 %
three	0	0	0	58	2	0	97 %
four	0	0	0	8	50	2	83 %
palm	0	0	0	0	1	59	98 %

5. Results

Metric	Measured
Recogniser accuracy, 360 synthetic frames	95.6 % (0 no-hand frames), 7.6 ms/frame
Position IK (200 random dexterous poses, LM + restarts)	200/200 converged, max 0.010 mm, mean 10.3 iterations, 3.0 ms
Full 6-DOF pose IK (100 poses, position + orientation)	89/100 converged; median 0.006 mm / 0.001 deg
Analytic planar IK	118/200 sampled targets reachable, max error 9.5e-14 mm
RMS joint jerk, scripted 12.8 s sequence	13512 -> 2523 deg/s ³ (81 % reduction)
Cartesian target jerk (raw gesture -> filtered)	82503 -> 20672 mm/s ³
Sequence tracking error (tool vs filtered target)	mean 2.9 mm, p95 20.8 mm (during the HOME jump), final 0.04 mm
Gesture read correctly in sequence (incl. 1-frame debounce lag)	95.8 % of 192 frames; modes visited: GRIP, HOME, JOG, POSITION, STOP, TOOL_TOGGLE
Peak joint speed in sequence	80 deg/s (servo cap 90-180 deg/s)

Gesture-controlled arm: smoothness before/after filtering

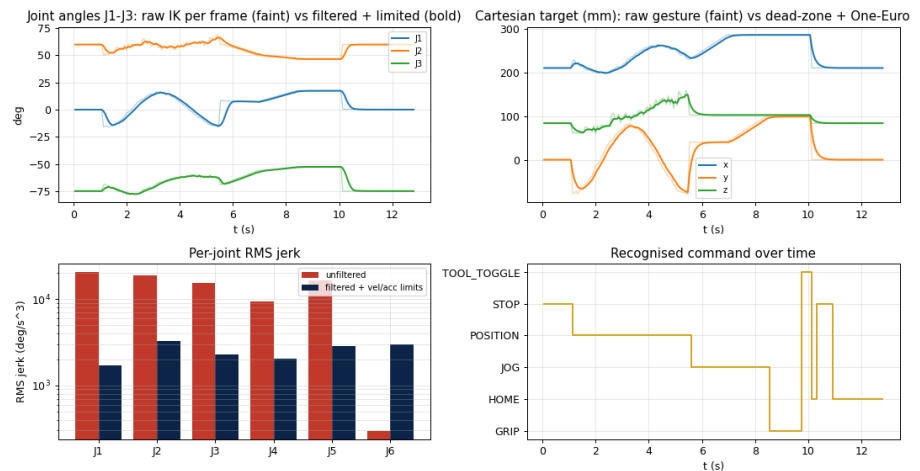


Figure 1 - smoothness.png: joint angles and Cartesian target, raw vs filtered; per-joint RMS jerk (log scale); recognised command timeline for the scripted sequence.

6. Example frames

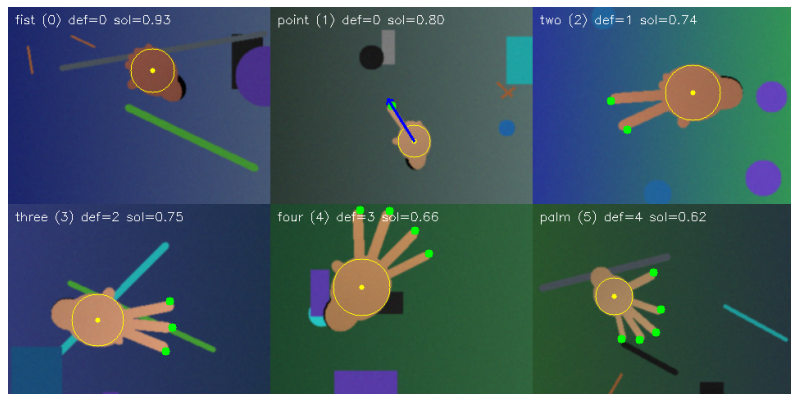


Figure 2 - gesture_examples.png: one rendered frame per class with the recogniser overlay (palm circle, fingertips, pointing arrow, valley count and solidity).

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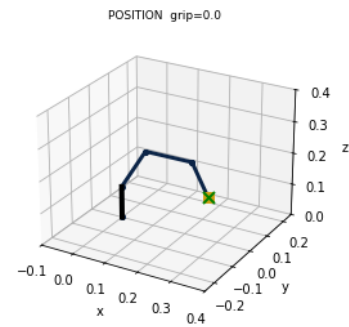
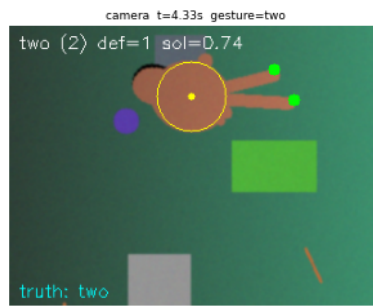


Figure 3 - a frame of gesture_arm.gif: camera view with overlay (left) and the 3-D arm tracking the filtered target (green x) in POSITION mode (right).

7. DH table and how to run

```
cd 07_Gesture_Controlled_Arm
./run_all.sh          # evaluation + IK benchmark + sequence + GIF, then 17 unit tests;
                        exit code
make sim              # -> sim/summary.json, confusion.png, smoothness.png,
                        gesture_arm.gif, sequence_log.csv
make test             # python3 -m unittest discover -s tests -v
make report           # docs/build_report.py -> tools/make_report.py -> docs/07_Report.pdf
python3 -m gesture_arm.webcam_demo --backend classical # on a laptop with a webcam
                        (not run here)
```

Joint	a (m)	alpha (rad)	d (m)	theta offset	limits (deg)	role
J1	0	+pi/2	0.100	0	-150 .. 150	base yaw
J2	0.150	0	0	0	-10 .. 110	shoulder pitch (elevation)
J3	0	+pi/2	0	+pi/2	-140 .. 10	elbow pitch; z3 = forearm axis
J4	0	-pi/2	0.140	0	-150 .. 150	forearm roll (forearm length in d4)
J5	0	+pi/2	0	0	-110 .. 110	wrist pitch
J6	0	0	0.110	0	-180 .. 180	tool roll (wrist + tool length)

8. Declared substitutions / not run here

- **MediaPipe:** src/gesture_arm/mediapipe_backend.py implements the same GestureBackend interface from the 21 hand landmarks (lazy import; the landmark-to-gesture part is unit-tested with synthetic landmarks). mediapipe is not installed here, so the backend itself was NOT run.
- **Webcam:** src/gesture_arm/webcam_demo.py (OpenCV capture -> backend -> loop -> optional serial output of the six servo angles) was NOT run - the container has no camera or display.
- **Webots:** worlds/gesture_arm.wbt (6 HingeJoints matching the DH geometry, gripper slider, webcam Camera) and worlds/controllers/gesture_arm_controller/ were NOT run - Webots is not available.
- **ROS 2 Humble:** ros2_ws/src/gesture_controlled_arm (node subscribing to the camera Image and publishing /gesture_arm/gesture, /gesture_arm/target, /joint_states, /arm_controller/joint_trajectory, /gesture_arm/gripper; launch file) compiles with py_compile but was NOT launched - no rclpy here.
- Everything else ran in this container with Python 3.11; all numbers above are read from sim/summary.json and sim/test_log.txt by docs/build_report.py.

9. Confusion matrix figure

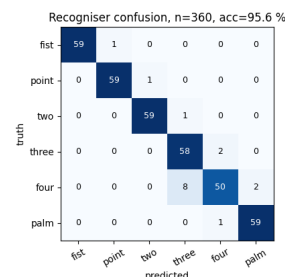


Figure 4 - confusion.png (same data as the table in section 4): all errors are adjacent finger counts.

10. Learnings / next steps

The recogniser only became reliable after I stopped trusting the raw defect count: the folded thumb and the wrist each create a plausible valley, and requiring both ends of a valley to be finger-length from the palm fixed most of the confusion. On the IK side the textbook DLS update failed on targets near the fully stretched boundary; adding LM step acceptance, freeing limit-locked joints and seeding from the analytic solution made position IK converge on every dexterous-workspace pose, while full-pose IK still fails on about a tenth of random orientations that the joint limits make hard to reach. Next: replace the fixed 4-way pointing quantisation with a continuous Cartesian jog, add a Kalman-smoothed hand-size estimate for depth, and run the webcam demo and Webots world on a laptop with a real hand.

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